

Task 2: Movie Rating Prediction

Project Overview

This project predicts movie ratings using a Machine Learning regression model. It analyzes movie-related information such as genre, director, and lead actors to estimate the IMDb rating of a movie.

Objective

Build a machine learning model to accurately predict movie ratings based on movie attributes.

Technologies Used

- Python
- Pandas
- Matplotlib
- Scikit-learn
- Google Colab

Workflow

- Import the required libraries.
- Load the movie dataset.
- Preprocess the data by handling missing values and encoding categorical variables.
- Split the dataset into training and testing sets.
- Train the Random Forest Regression model.
- Evaluate the model using regression metrics.
- Visualize the movie rating distribution and genre statistics.
- Predict the rating of a sample movie.

Result

The model successfully predicts movie ratings using the Random Forest Regression algorithm. Performance is evaluated using MAE, MSE, and R^2 Score, while visualizations provide insights into the movie dataset and prediction results.